MATE70020 Problem Set 2

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10 November 2026

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(a) Why the $6 \times 6 \times 1$ grid has no k-points along z

Quantum ESPRESSO uses periodic boundary conditions in all three directions, so BN.in repeats the single hBN sheet along z with period $c = 20.0$ Å ($\vec{a}_3 = c\,\hat{\vec{z}}$). Each sheet is therefore separated from its (artificial) periodic images by about 20 Å of vacuum, just like the isolated N_2 molecule in a box in Lecture 3. The Kohn–Sham orbitals of neighbouring sheets do not overlap across this vacuum, so the occupied KS energies $\varepsilon_{n\vec{k}}$ do not depend on k_z: the bands are flat along k_z and every k_z contributes the same amount to the density $n(\vec{r}) = \sum_n \sum_{\vec{k}} |\psi_{n\vec{k}}(\vec{r})|^2$ (Lecture 4). A single k_z value ($k_z = 0$) therefore samples the Brillouin zone exactly along z, and the k-grid only has to be converged in the plane of the sheet, where the material is really periodic. Equivalently, the zone is very thin along z: $|\vec{b}_3| = 2\pi/c = 0.314$ Å^{-1}, 9.2 times shorter than $|\vec{b}_1| = 2.897$ Å^{-1} (part b). Check: the occupied KS energies at $k_z = \pi/c$ (at Γ and K) differ from those at $k_z = 0$ by less than 0.01 eV, whereas band 4 disperses by 3.6 eV in the plane.

(b) Primitive lattice vectors and reciprocal lattice vectors

For `ibrav = 4` (hexagonal) the INPUT_PW documentation [1] gives, in Cartesian coordinates, $\vec{a}_1 = a(1,0,0)$, $\vec{a}_2 = a(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$, $\vec{a}_3 = a(0,0,\frac{c}{a})$. In BN.in, `A` and `C` are lengths in Å [1], so $a = 2.504$ Å and $c = 20.0$ Å:

$$\vec{a}_1 = (2.504,\ 0,\ 0)\ \text{Å}, \qquad \vec{a}_2 = (-1.252,\ 2.169,\ 0)\ \text{Å}, \qquad \vec{a}_3 = (0,\ 0,\ 20.0)\ \text{Å}.$$

The reciprocal vectors follow from $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$, i.e. $\vec{b}_1 = 2\pi\,\vec{a}_2 \times \vec{a}_3 / V$ (cyclic), with $V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)$ (Lecture 4):

$$\vec{a}_2 \times \vec{a}_3 = ac\,(\tfrac{\sqrt{3}}{2}, \tfrac{1}{2}, 0), \quad \vec{a}_3 \times \vec{a}_1 = ac\,(0,1,0), \quad \vec{a}_1 \times \vec{a}_2 = a^2\,(0,0,\tfrac{\sqrt{3}}{2}), \quad V = \tfrac{\sqrt{3}}{2}a^2 c = 108.60\ \text{Å}^3,$$

$$\vec{b}_1 = \frac{2\pi}{a}\Big(1, \tfrac{1}{\sqrt{3}}, 0\Big), \qquad \vec{b}_2 = \frac{2\pi}{a}\Big(0, \tfrac{2}{\sqrt{3}}, 0\Big), \qquad \vec{b}_3 = \frac{2\pi}{c}\,(0,0,1).$$

With $2\pi/a = 2.5093$ Å^{-1} and $2\pi/c = 0.31416$ Å^{-1}:

$$\vec{b}_1 = (2.5093,\ 1.4487,\ 0)\ \text{Å}^{-1}, \qquad \vec{b}_2 = (0,\ 2.8974,\ 0)\ \text{Å}^{-1}, \qquad \vec{b}_3 = (0,\ 0,\ 0.3142)\ \text{Å}^{-1},$$

so $|\vec{b}_1| = |\vec{b}_2| = 4\pi/(\sqrt{3}a) = 2.8974$ Å^{-1} at 60° to each other (a hexagonal reciprocal lattice). Check: e.g. $\vec{a}_2 \cdot \vec{b}_1 = 2\pi(-\frac{1}{2} + \frac{\sqrt{3}}{2}\frac{1}{\sqrt{3}}) = 0$, $\vec{a}_2 \cdot \vec{b}_2 = 2\pi$; numerically $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ to 10^{-15}. In units of $2\pi/a$ these are $(1, 0.57735, 0)$, $(0, 1.15470, 0)$ and $(0, 0, 0.12520)$, identical to the reciprocal axes printed by `pw.x`.

(c) K and M in terms of the primitive reciprocal lattice vectors

The first Brillouin zone is the region closer to $\vec{G} = 0$ than to any other $\vec{G}$ (Lecture 4). Its boundary is formed by the perpendicular bisectors of the six shortest $\vec{G}$: $\pm\vec{b}_1$, $\pm\vec{b}_2$, $\pm(\vec{b}_2 - \vec{b}_1)$, all of length $|\vec{b}_1|$.

M (middle of a side). The side belonging to $\vec{b}_1$ is its perpendicular bisector, and its midpoint is where it crosses the line from Γ to $\vec{b}_1$: $\vec{k}_M = \frac{1}{2}\vec{b}_1$, i.e. crystal coordinates $(\frac{1}{2}, 0, 0)$.

K (corner). A corner is where the sides belonging to $\vec{b}_1$ and $\vec{b}_2$ meet, so it is equidistant from Γ, $\vec{b}_1$ and $\vec{b}_2$. Write $\vec{k}_K = x\vec{b}_1 + y\vec{b}_2$ and use $|\vec{b}_1|^2 = |\vec{b}_2|^2 \equiv B^2$, $\vec{b}_1 \cdot \vec{b}_2 = B^2/2$:

$$|\vec{k}_K| = |\vec{k}_K - \vec{b}_1| \Rightarrow 2\vec{k}_K \cdot \vec{b}_1 = B^2 \Rightarrow 2x + y = 1, \qquad |\vec{k}_K| = |\vec{k}_K - \vec{b}_2| \Rightarrow x + 2y = 1,$$

so $x = y = \frac{1}{3}$: $\vec{k}_K = \frac{1}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2$, crystal coordinates $(\frac{1}{3}, \frac{1}{3}, 0)$. Γ is $(0, 0, 0)$.

Check in Cartesian coordinates.

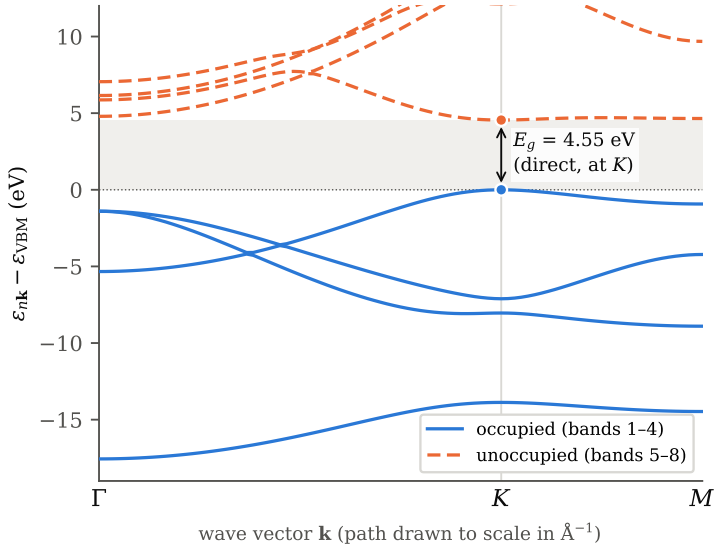
$$\vec{k}_K = \tfrac{2\pi}{a}(\tfrac{1}{3}, \tfrac{1}{\sqrt{3}}, 0) = (0.8364,\ 1.4487,\ 0)\ \text{Å}^{-1}, \qquad \vec{k}_M = \tfrac{2\pi}{a}(\tfrac{1}{2}, \tfrac{1}{2\sqrt{3}}, 0) = (1.2546,\ 0.7244,\ 0)\ \text{Å}^{-1},$$

giving $|\Gamma K| = 4\pi/(3a) = 1.6728$ Å^{-1}, $|\Gamma M| = 2\pi/(\sqrt{3}a) = 1.4487$ Å^{-1} and $|KM| = 2\pi/(3a) = 0.8364$ Å^{-1}. For a regular hexagon the corner distance equals the side length s, and the distance to the middle of a side is $\frac{\sqrt{3}}{2}s$. Here $|\Gamma M|/|\Gamma K| = \sqrt{3}/2$ and $|KM| = \frac{1}{2}|\Gamma K|$, so M is the midpoint of the side starting at K (the side running to $K' = (\frac{2}{3}, -\frac{1}{3}, 0)$), as in the sketch on the problem sheet. A numerical search over $\vec{G}$ also confirms that K is equidistant from 3 nearest $\vec{G}$ (a corner) and M from 2 (a side midpoint).

(d) Band structure of monolayer hBN

Method. Quantum ESPRESSO 7.6 [3], LDA (Perdew–Zunger) pseudopotentials as supplied. (1) SCF with BN.in *unchanged* ($E_{\text{cut}} = 60$ Ry, $6\times6\times1$ grid, `nbnd = 8`). It converged in 7 iterations (estimated SCF accuracy 3×10^{-7} Ry, below the default 10^{-6} Ry), with $E_{\text{tot}} = -25.77488$ Ry. (2) A non-self-consistent run on that SCF density. The only changes to BN.in are `calculation = 'bands'` and `K_POINTS crystal_b` with Γ $(0,0,0)$ 60, K $(\frac{1}{3},\frac{1}{3},0)$ 30, M $(\frac{1}{2},0,0)$ 1. This gives 91 k-points with equal spacing (0.028 Å^{-1}, since $|\Gamma K| = 2|KM|$). (3) Post-processing with `bands.x`.

Occupied bands. The pseudopotentials have $Z_{\text{val}} = 3$ (B) and 5 (N), so there are 8 valence electrons per cell. In a spin-unpolarised calculation each band holds 2 electrons, so bands 1–4 are occupied and bands 5–8 are empty.



(b) $a = 2.504$ Å, $c = 20.0$ Å	
$\vec{b}_1$	(2.509, 1.449, 0) Å^{-1}
$\vec{b}_2$	(0, 2.897, 0) Å^{-1}
$\vec{b}_3$	(0, 0, 0.314) Å^{-1}
(c) crystal coordinates	
Γ, K, M	$(0,0,0)$, $(\frac{1}{3},\frac{1}{3},0)$, $(\frac{1}{2},0,0)$
$\lvert\Gamma K\rvert$, $\lvert\Gamma M\rvert$	1.673 Å^{-1}, 1.449 Å^{-1}
(d) 8 valence e^-, 4 occupied bands	
VBM, CBM	both at K
E_g (LDA)	4.55 eV, direct at K
E_g (expt. [2])	6.1 eV, direct
difference	−1.55 eV (−25%)

Figure 1: Left: LDA Kohn–Sham band structure of monolayer hBN along Γ–K–M (path to scale: $|\Gamma K| = 1.673$ Å^{-1}, $|KM| = 0.836$ Å^{-1}). Energies are relative to the valence-band maximum. Solid blue: occupied bands 1–4; dashed orange: unoccupied bands 5–8; shading: band gap. Right: numerical results for (b)–(d).

Result. The valence-band maximum (top of band 4) and the conduction-band minimum (bottom of band 5) are both at K. **hBN therefore has a direct band gap at K, $E_g = \varepsilon_{5,K} - \varepsilon_{4,K} = 0.207 - (-4.343) = 4.55$ eV.** Elsewhere on the path the gap is larger: 5.58 eV at M and 6.19 eV at Γ. A 3× denser path (271 k-points) and a fine path within 0.17 Å^{-1} of K (201 k-points) give the same location and the same gap (4.55 eV).

Comparison with experiment. The measured gap of monolayer hBN is 6.1 eV, and it is also direct [2]. LDA reproduces the direct character but underestimates the gap by 1.55 eV ($\approx 25\%$). This is the known failure discussed in Lecture 3: LDA (and GGAs) work well for structural, elastic and vibrational properties but are inaccurate for band gaps of solids, and semi-local functionals systematically underestimate them. Two reasons are relevant here. First, the KS energies are eigenvalues

of the fictitious non-interacting system, not true excitation energies (Lecture 2). Second, LDA takes ϵ_{xc} from the homogeneous electron gas, evaluated locally. Its approximate exchange does not cancel the spurious interaction of each electron with its own density, which is contained in the Hartree term. Exact (Hartree–Fock) exchange cancels this self-interaction. Hybrid functionals such as PBE0 replace 25% of the semi-local exchange by Hartree–Fock exchange, which opens the gap towards experiment, but they are roughly an order of magnitude more expensive.

References

[1] Quantum ESPRESSO, *Input File Description, Program pw.x* (INPUT_PW): entries `ibrav`, `A/C`, `K_POINTS`; `https://www.quantum-espresso.org/Doc/INPUT_PW.html` (accessed 10 Nov 2026).
[2] C. Elias, P. Valvin, T. Pelini, A. Summerfield, C. J. Mellor, T. S. Cheng, L. Eaves, C. T. Foxon, P. H. Beton, S. V. Novikov, B. Gil, G. Cassabois, "Direct band-gap crossover in epitaxial monolayer boron nitride", *Nat. Commun.* **10**, 2639 (2019), doi:10.1038/s41467-019-10610-5.
[3] P. Giannozzi *et al.*, *J. Phys.: Condens. Matter* **29**, 465901 (2017).